

Market & Customer Segmentation Analysis

Unsupervised learning on Instacart transaction behavior

Problem → data → behavior → models → segments → decisions

3.4M

orders analyzed

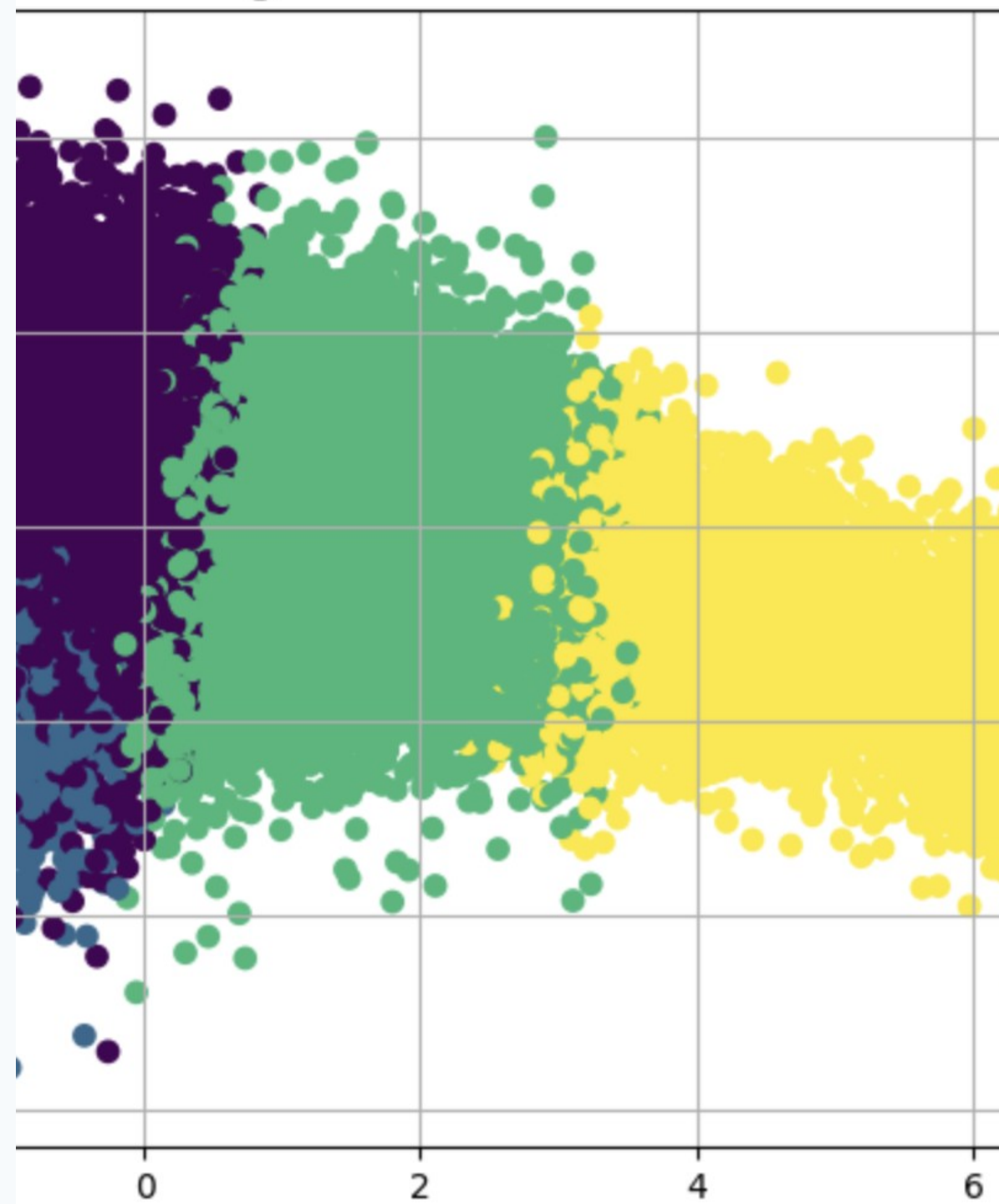
200K+

customers profiled

32M+

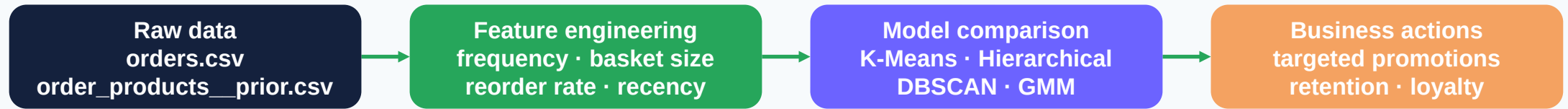
order-product rows

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Building a customer-level segmentation pipeline

Convert transaction logs into interpretable behavioral profiles, then compare clustering approaches.

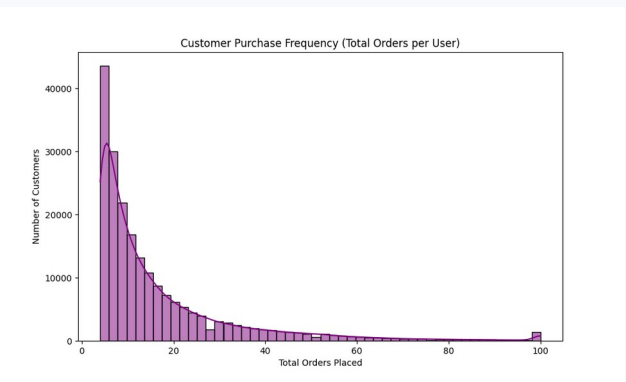


Behavioral signals used for segmentation

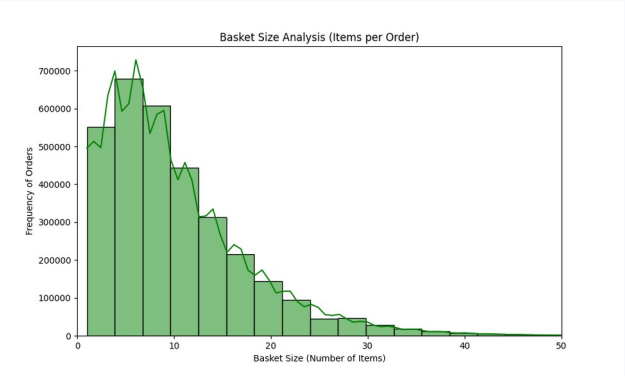
EDA showed non-uniform shopping patterns, long-tailed customer activity, and meaningful reorder behavior.



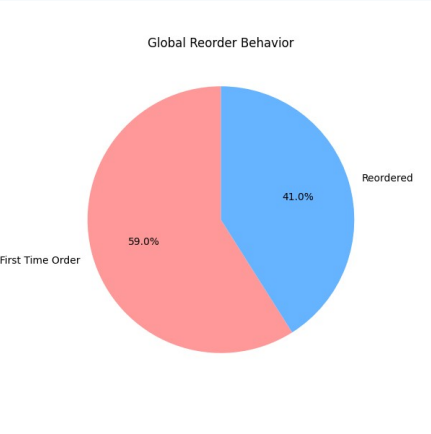
Peak windows



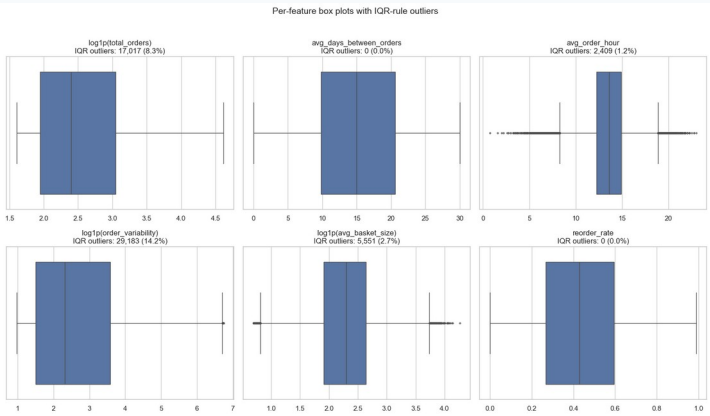
Purchase frequency



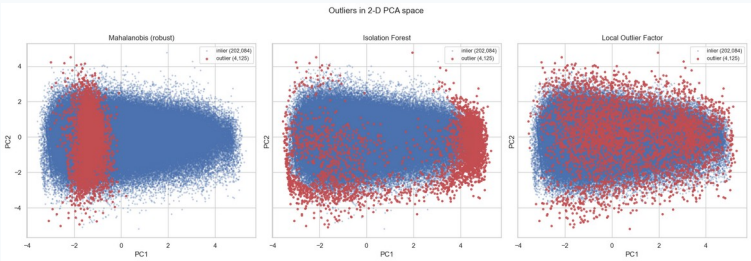
Basket size



Reorder loyalty



Outlier profile

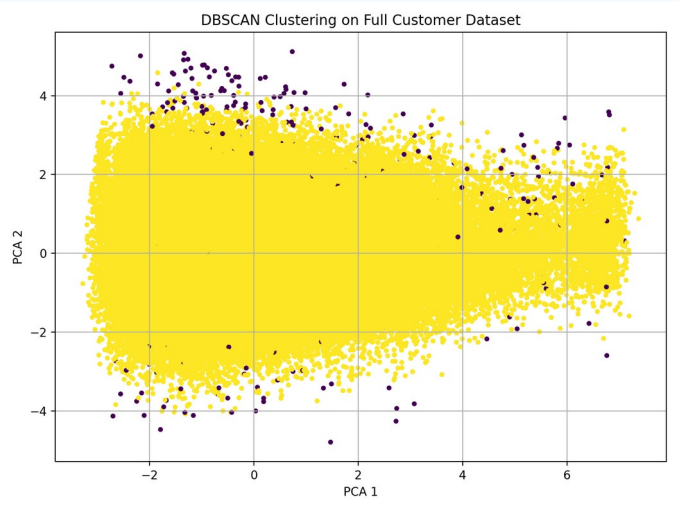
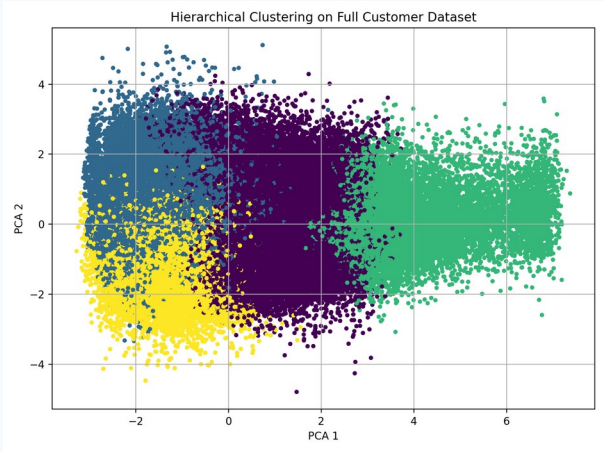
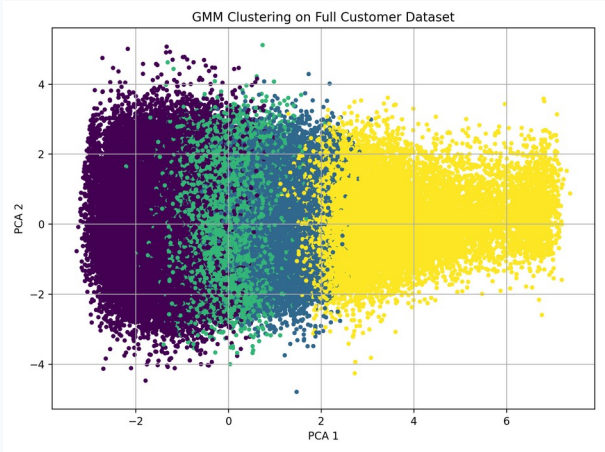


Detector agreement

Model comparison: four clusters are interpretable

K-Means, Hierarchical, and GMM converged on useful segments; DBSCAN mainly served as an outlier check.

Criterion	K-Means	Hierarchical	DBSCAN	GMM
Clusters found	4	4	1 + noise	4
Dataset size	188,816 (outliers removed)	206,209 (full, HPC)	206,209 (full, HPC)	206,209 (full, HPC)
Silhouette score	0.2574	Visual only	N/A	~0.25
Soft membership	No	No	No	Yes
Outlier handling	Pre-removed	Not built-in	Yes (345 pts)	Partial
Log transform used	Yes (orders+basket)	No	No	No
Recommended role	Baseline + EDA	K validation	Outlier monitor	Primary model



K-Means baseline
K = 4 offered the best balance of interpretability and separation.
Silhouette score: 0.2574.

Hierarchical validation
Dendrogram and PCA plots supported four natural customer groups, but full dendrogram visualization was costly.

GMM as primary model
Soft membership captures overlapping customer behaviors better than hard segment boundaries.

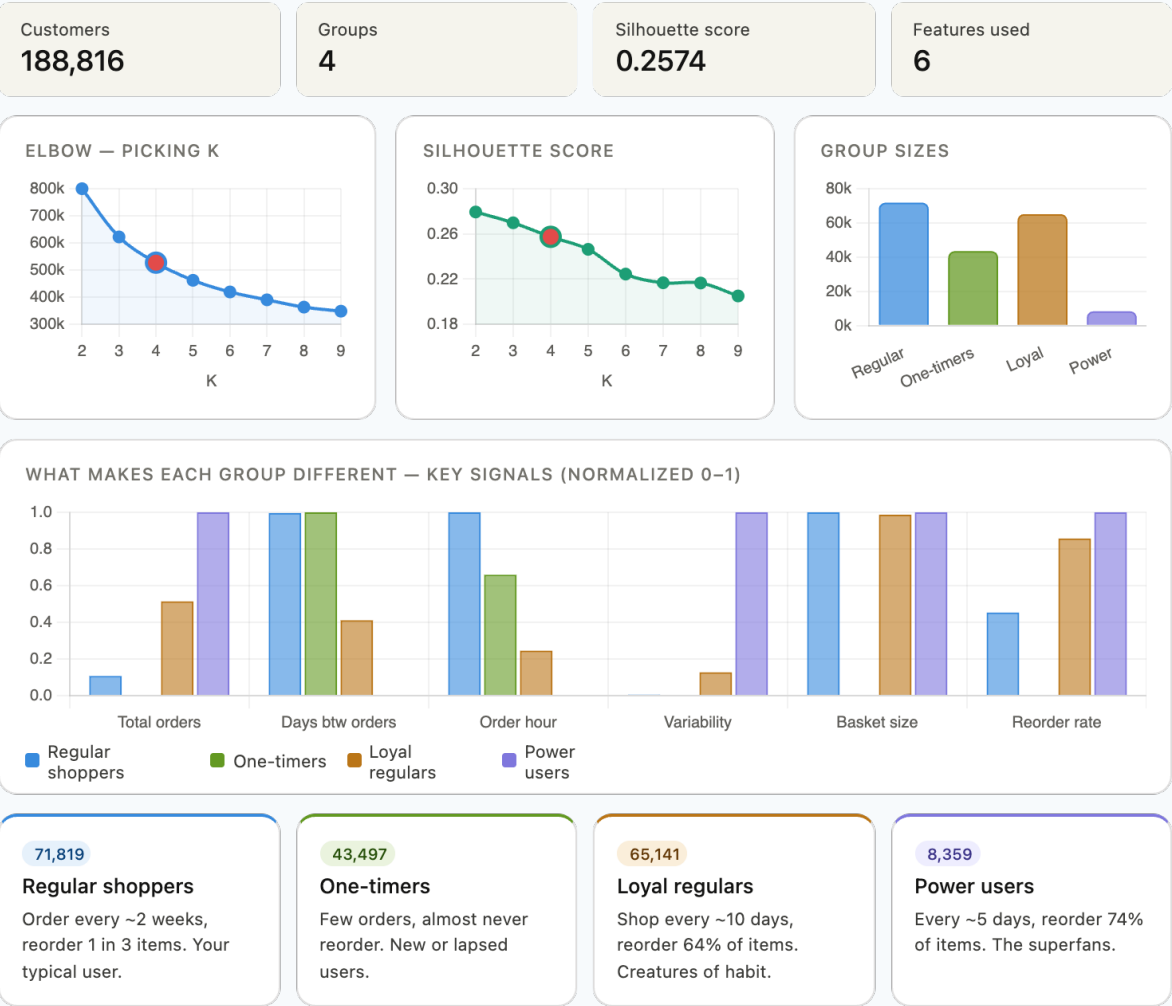
Customer segments translate into marketing actions

The models consistently identified four meaningful behavioral groups, with late-hour shoppers as a sub-group.

CUSTOMER SEGMENTS (K-MEANS, HIERARCHICAL, GMM)			
Cluster	Segment Name	Behavior Description	Business Insight
 Cluster 0	Moderate Shoppers	Customers who shop regularly with moderate purchase frequency and average reorder behavior.	 Target with personalized promotions and product recommendations to increase engagement and spending.
 Cluster 1	Low-Engagement Users	Customers with low purchase frequency and minimal reorder activity.	 Use retention strategies such as discounts, onboarding campaigns, and reminder notifications to re-engage these users.
 Cluster 2	Loyal Customers	Customers who frequently reorder the same products and show consistent purchasing patterns.	 Ideal for loyalty programs , subscriptions , and personalized offers to maintain long-term engagement.
 Cluster 3	High-Value / Power Users	Customers with high purchase frequency, larger basket sizes, and strong reorder behavior.	 Prioritize for premium services , exclusive deals , and targeted marketing strategies to maximize revenue.
MODEL-SPECIFIC INSIGHTS			
Model	Observation	Business Insight	
 DBSCAN	DBSCAN did not produce multiple clusters and instead grouped most customers into a single segment with a few outliers.		Indicates that customer behavior is continuous rather than forming clearly separable density-based groups, making DBSCAN less suitable for segmentation .
 GMM	GMM provided flexible clustering with overlapping segments, allowing customers to belong partially to multiple groups.		Enables businesses to design hybrid marketing strategies , where customers can be targeted based on multiple behavioral characteristics.

Decision summary and next steps

The strongest story is not sharp separation; it is behavior that changes gradually across overlapping customer profiles.



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Final conclusions

- 1 Customer-level features made the large transaction dataset clusterable and interpretable.
- 2 K-Means and Hierarchical Clustering validated the existence of four useful segments.
- 3 DBSCAN was not suitable for segmentation because it collapsed most users into one density group.
- 4 GMM is the recommended primary model because soft membership fits overlapping retail behavior.

Next steps

Attach segment labels to campaigns, run A/B tests by segment, and add product/category features to refine high-value and loyal customer profiles.

Appendix

Why customer-level aggregation?

Clustering millions of disconnected transaction rows is noisy. A single profile per user makes patterns comparable across customers.

```
train_matrix.head()
```

[8] ✓ 0.0s Python

	order_id	user_id	order_number	product_id	target	user_total_orders	user_avg_days_between_orders	user_median_day
0	1187899	1	11	196	1	10	19.555556	
1	1187899	1	11	10258	1	10	19.555556	
2	1187899	1	11	10326	0	10	19.555556	
3	1187899	1	11	12427	0	10	19.555556	
4	1187899	1	11	13032	1	10	19.555556	

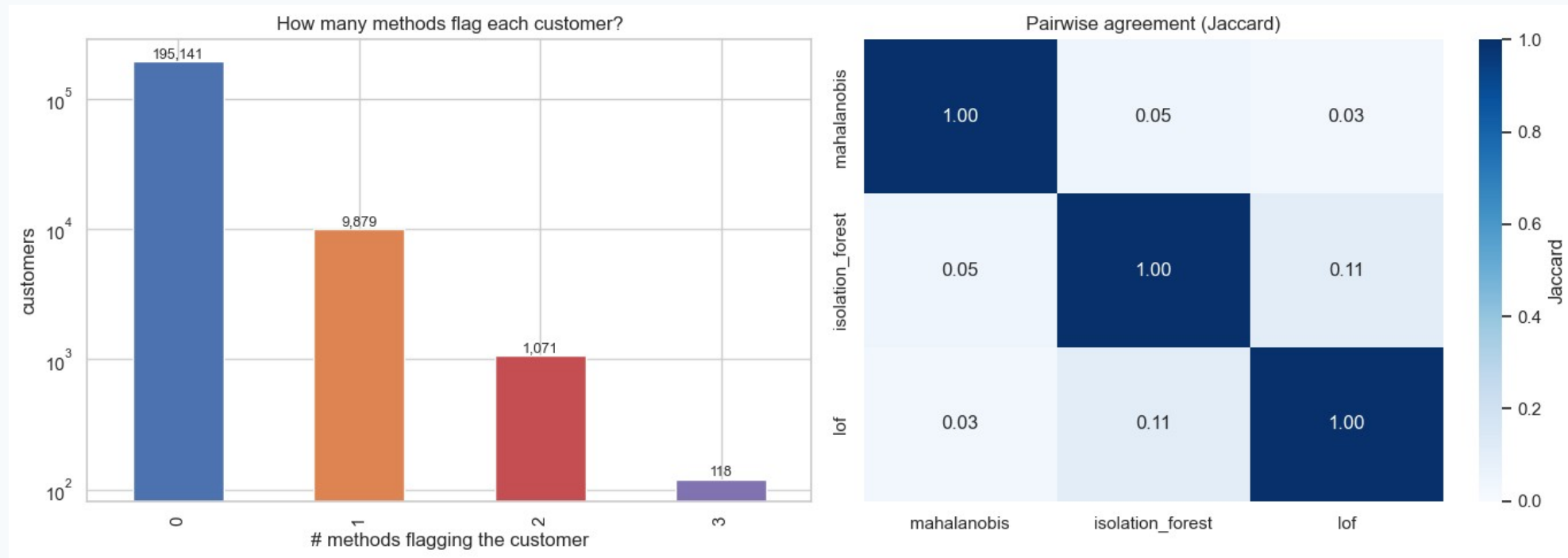
5 rows x 49 columns

```
#train_matrix.dataframe()  
train_matrix.to_csv("../data/train_matrix.csv")
```

[9] ✓ 1m 57.1s Python

Why multiple algorithms?

Retail behavior has outliers, overlap, and irregular purchase patterns. Comparing methods reduces model-specific bias.



Final modeling choice

The full engineered feature set was preserved after PCA exploration to avoid losing behavioral detail.

4

Clusters found

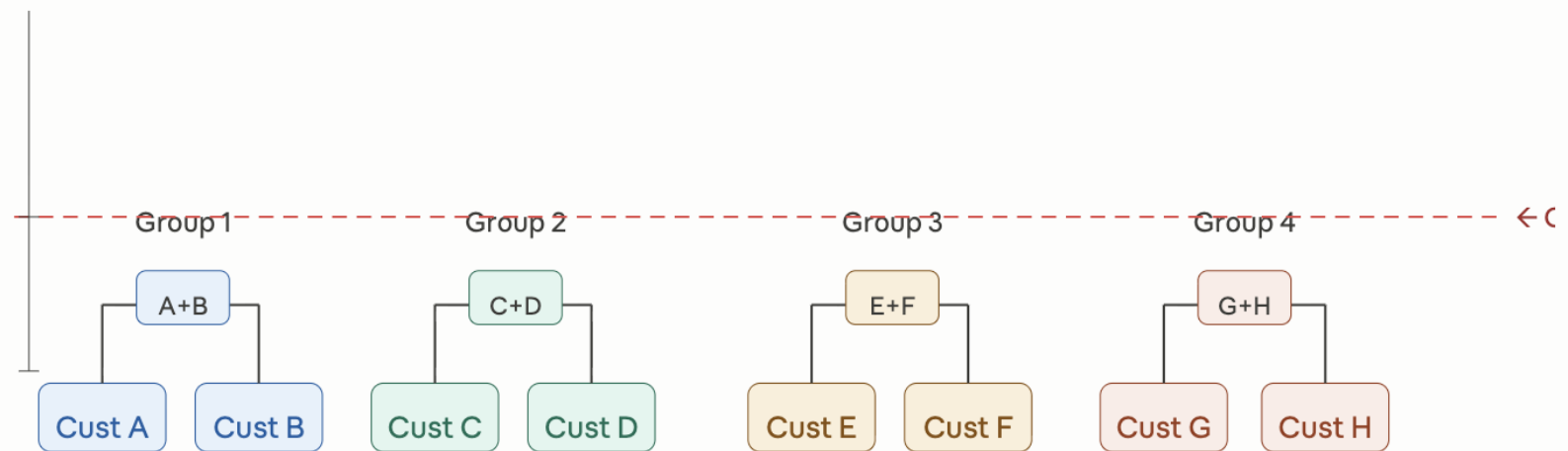
HPC

Computing needed

Dendrogram

Key output

Higher = more different



Verdict: Worked well. The dendrogram clearly showed 4 natural groups with a large gap before the next merge — strong evidence that K=4 is the right number. Used as the benchmark for the other models.

Customer segments translate into marketing actions

The models consistently identified four meaningful behavioral groups, with late-hour shoppers as a sub-group.

Although each model assigns its own cluster labels, the underlying behavioural patterns are consistent across K-Means, Hierarchical Clustering, and GMM. The following unified naming framework is used throughout this section to refer to equivalent segments regardless of which model produced them:

Behaviour	Unified Segment Name
High frequency, high reorder (~5 days, ~73%)	Power Users / Champions
Medium-high frequency, high reorder (~10-12 days, ~55-64%)	Loyal Regulars
Low-medium frequency, low-medium reorder (~18-19 days, ~28-34%)	Casual / Infrequent Shoppers
Near-zero reorder, very low engagement	New / Inactive Users